

Carnegie Mellon University
MSCF Program

Matrices & Linear Algebra – Part 2 (MSCF Math Prep)

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Determinants

Let $A \in M^{n \times n}$ be given.

If $n = 1$, so that $A = (A_{11})$, we have $\det(A) = A_{11}$.

If $n = 2$, so that

$$A = \begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix},$$

then we have

$$\det(A) = A_{11}A_{22} - A_{12}A_{21}.$$

The absolute value of $\det(A)$ is equal to the area of the parallelogram determined by the vectors forming the columns of A . Sometimes the determinant of a matrix A is indicated by putting bars around A , i.e.

$$|A| = \det(A).$$

A General Formula

For a general $n \times n$ matrix A , let $\mathcal{A}^{(ij)}$ denote the $(n-1) \times (n-1)$ matrix obtained by deleting row i and column j from A . We can define the determinant of A by

$$\det(A) = \sum_{j=1}^n (-1)^{1+j} A_{1j} \det(\mathcal{A}^{(1j)}).$$

This way of computing a determinant is referred to as *cofactor expansion along the first row*.

Let's see what this formula says in the 2×2 case:

$$A = \begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix},$$

$$\mathcal{A}^{(11)} = A_{22}, \quad \mathcal{A}^{(12)} = A_{21}, \quad \mathcal{A}^{(21)} = A_{12}, \quad \mathcal{A}^{(22)} = A_{11}.$$

We see that

$$\begin{aligned} \det(A) &= \sum_{j=1}^2 (-1)^{1+j} A_{1j} \det(\mathcal{A}^{(1j)}) \\ &= A_{11} A_{22} - A_{12} A_{21}. \end{aligned}$$

We can compute a determinant by using cofactor expansion along any row (or any column). Choose any $i \in \{1, 2, 3, \dots, n\}$. Then we have

$$\det(A) = \sum_{j=1}^n (-1)^{i+j} A_{ij} \det(\mathcal{A}^{(ij)}).$$

Remark: It is true that $\det(A) = \det(A^\top)$, and this shows that we can compute $\det(A)$ by using cofactor expansion along any column.

When $n = 3$, $|\det(A)|$ is equal to the volume of the parallelepiped determined by the columns of A (and this is the same as the volume of the parallelepiped determined by the rows of A).

In n dimensions the volume of the parallelotope determined by the columns of A is equal to $|\det(A)|$.

Remark: Cofactor expansion is **NOT** an efficient way to actually compute determinants when n is large.

Some Important Properties of Determinants

Let $A, B \in M^{n \times n}$ and a scalar α be given.

1. $\det(A) = \det(A^\top)$
2. $\det(\alpha A) = \alpha^n \det(A)$.
3. $\det(AB) = \det(A) \cdot \det(B)$
4. If $D \in M^{n \times n}$ is diagonal then

$$\det(D) = D_{11} D_{22} D_{33} \cdots D_{nn}.$$

5. $\det(A) \neq 0$ if and only if the columns of A are linearly independent.

Properties of Determinants (Continued)

- 6. $\det(A) \neq 0$ if and only if $\text{rank}(A) = n$.
- 7. $\det(A) \neq 0$ if and only if $\mathcal{N}(A) = \{0\}$.
- 8. A is invertible if and only if $\det(A) \neq 0$.
- 9. If A is invertible then

$$\det(A^{-1}) = \frac{1}{\det(A)}.$$

Example: $\det \begin{pmatrix} 1 & 3 \\ -5 & 2 \end{pmatrix} = 1(2) - 3(-5) = 2 + 15 = 17$

Example: Let $A = \begin{pmatrix} 1 & 3 & 5 \\ 2 & -1 & 0 \\ 1 & 2 & 4 \end{pmatrix}$. Let's compute $\det(A)$ by using cofactor expansion along the second row.

$$\begin{aligned} \det(A) &= (-1)^3(2)\det \begin{pmatrix} 3 & 5 \\ 2 & 4 \end{pmatrix} + (-1)^4(-1)\det \begin{pmatrix} 1 & 5 \\ 1 & 4 \end{pmatrix} + 0 \\ &= -2(12 - 10) - 1(4 - 5) \\ &= -4 - (-1) = -3. \end{aligned}$$

Explicit Formula for A^{-1}

Let $A \in M^{n \times n}$ be given. The *adjugate* of A is the $n \times n$ matrix defined by

$$[\text{adj}(A)]_{ij} = (-1)^{i+j} \det(A^{(ji)}).$$

Fact: If $\det(A) \neq 0$ then

$$A^{-1} = \frac{\text{adj}(A)}{\det(A)}.$$

When n is large, this is **NOT** an efficient way to actually compute A^{-1} . However, it is sometimes useful to compute A^{-1} when n is 2 or 3. (It is also sometimes useful from a theoretical perspective.) Let's do a 2×2 example.

Example. Let $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$ and observe that $\det(A) = -2 \neq 0$.

$$\mathcal{A}^{(11)} = 4, \quad \mathcal{A}^{(12)} = 3, \quad \mathcal{A}^{(21)} = 2, \quad \mathcal{A}^{(22)} = 1,$$

$$[\operatorname{adj}(A)]_{11} = (-1)^2 \cdot 4 = 4, \quad [\operatorname{adj}(A)]_{12} = (-1)^3 \cdot 2 = -2,$$

$$[\operatorname{adj}(A)]_{21} = (-1)^3 \cdot 3 = -3, \quad [\operatorname{adj}(A)]_{22} = (-1)^4 \cdot 1 = 1.$$

$$A^{-1} = \frac{1}{-2} \begin{pmatrix} 4 & -2 \\ -3 & 1 \end{pmatrix} = \begin{pmatrix} -2 & 1 \\ \frac{3}{2} & -\frac{1}{2} \end{pmatrix}$$

A Remark on Skew Matrices

An $n \times n$ matrix B is said to be *skew symmetric* provided that $B^T = -B$.

It follows that

$$\det(B) = (-1)^n \det(B)$$

if B is skew symmetric. In particular, $\det(B) = 0$ if n is odd and B is skew symmetric.

Upper and Lower Triangular Matrices

An $n \times n$ matrix L is said to be *lower triangular* provided that $L_{ij} = 0$ for all $i, j = 1, 2, \dots, n$ with $i < j$.

An $n \times n$ matrix U is said to be *upper triangular* provided that for $U_{ij} = 0$ for all $i, j = 1, 2, \dots, n$ with $i > j$.

Observe that a diagonal matrix is both upper and lower triangular.

Fact: Let $T \in M^{n \times n}$ be given and assume that T is upper triangular or lower triangular then

$$\det(T) = T_{11} T_{22} \cdots T_{nn}.$$

Row Operations and Determinants

Sometimes a determinant can be computed efficiently by using elementary row operations to obtain a matrix that is upper or lower triangular. In order to implement this procedure, we need to understand how each of the elementary row operations influences determinants.

1. If B is obtained by multiplying one row of A by a scalar α then $\det(B) = \alpha \det(A)$.
2. If C is obtained by interchanging two rows of A then $\det(C) = -\det(A)$.
3. If E is obtained by adding a scalar multiple of one row of A to another row then $\det(E) = \det(A)$.

Trace of a Matrix

Let $A \in M^{n \times n}$ be given. The *trace* of A is defined by

$$\operatorname{tr}(A) = \sum_{i=1}^n A_{ii}.$$

Let $A, B \in M^{n \times n}$ and a scalar α be given.

1. $\operatorname{tr}(A + B) = \operatorname{tr}(A) + \operatorname{tr}(B)$
2. $\operatorname{tr}(\alpha A) = \alpha \operatorname{tr}(A)$
3. $\operatorname{tr}(AB) = \operatorname{tr}(BA)$

Eigenvalues

Let A be an $n \times n$ matrix with real entries. We will allow scalars and column vectors (and certain matrices constructed from A) to have complex entries.

Definition: A scalar λ is said to be an *eigenvalue* for A provided there exists a nonzero $x \in V^n$ such that $Ax = \lambda x$. (Any nonzero x such that $Ax = \lambda x$ is called an *eigenvector* for A associated with the eigenvalue λ .)

$$Ax = \lambda x = \lambda(Ix) = (\lambda I)x$$

Here the matrix λI is allowed to have complex entries.

$$(A - \lambda I)x = 0 \quad (\text{want nonzero } x)$$

$$\mathcal{N}(A - \lambda I) \neq \{0\}$$

$$\det(A - \lambda I) = 0$$

Note that $\det(A - \lambda I)$ is a polynomial of degree n with leading coefficient $(-1)^n$. It is called the *characteristic polynomial* for A .

The equation $\det(A - \lambda I) = 0$ is called the *characteristic equation* for A .

By the fundamental theorem of algebra there exist complex scalars $\lambda_1, \lambda_2, \dots, \lambda_n$ such that

$$\det(A - \lambda I) = (-1)^n (\lambda - \lambda_1)(\lambda - \lambda_2) \cdots (\lambda - \lambda_n).$$

The scalars $\lambda_1, \lambda_2, \dots, \lambda_n$ are eigenvalues of A . (There could be repeats on this list. In other words, these n values need not be distinct.)

Let A be an $n \times n$ matrix and let $\lambda_1, \lambda_2, \dots, \lambda_n$ be the eigenvalues (with each distinct eigenvalue appearing on the list according to its multiplicity as a root of the characteristic equation). Then we have

- ▶ $\sum_{i=1}^n \lambda_i = \text{tr}(A)$
- ▶ $\prod_{i=1}^n \lambda_i = \det(A)$

The order in which we list the eigenvalues does not matter, but the number of times each value appears on the list is crucial.

Example

Suppose that the characteristic polynomial for A is

$$(\lambda - 1)(\lambda - 3)^2(\lambda + 2)^3.$$

Then we may take

$$\lambda_1 = 1, \lambda_2 = \lambda_3 = 3, \lambda_4 = \lambda_5 = \lambda_6 = -2.$$

Therefore we have

$$\operatorname{tr}(A) = 1 + 3 + 3 - 2 - 2 - 2 = 1,$$

and

$$\det(A) = 1(3)^2(-2)^3 = -72.$$

Similar Matrices

Let $A, B \in M^{n \times n}$ be given. We say that A and B are *similar* provided that there exists an invertible $T \in M^{n \times n}$ such that

$$B = T^{-1}AT.$$

(Notice that $A = S^{-1}BS$ where $S = T^{-1}$.)

Assuming that A and B are similar, let's check out $\det(B)$.

$$\begin{aligned}\det(B) &= \det(T^{-1}) \cdot \det(AT) \\ &= \det(T^{-1}) \cdot \det(A) \cdot \det(T) \\ &= (\det(T))^{-1} \cdot \det(A) \cdot \det(T) = \det(A)\end{aligned}$$

Properties of Similar Matrices

Suppose that $A, B \in M^{n \times n}$ are similar. Then

1. $\det(B) = \det(A)$
2. $\operatorname{tr}(B) = \operatorname{tr}(A)$
3. A and B have the same eigenvalues.

To see why item 3 is true, observe that if $B = T^{-1}AT$ then

$$B - \lambda I = T^{-1}(A - \lambda I)T,$$

so that

$$\det(B - \lambda I) = \det(A - \lambda I).$$

Real Symmetric Matrices

Theorem: Assume that the entries of A are real and that $A = A^\top$. Then all eigenvalues of A are real. Moreover there exist eigenvectors $e^{(1)}, e^{(2)}, \dots, e^{(n)}$ having real entries such that

$$(e^{(i)})^\top e^{(j)} = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases}.$$

Let Q be the $n \times n$ matrix whose i^{th} column is $e^{(i)}$, i.e.

$$Q_{ij} = e_i^{(j)}.$$

Then we have

$$(Q^\top Q)_{ij} = (e^{(i)})^\top e^{(j)}.$$

Therefore we have

$$Q^{\top}Q = QQ^{\top} = I,$$

so that $Q^{\top} = Q^{-1}$. We also have

$$Q^{\top}AQ = \begin{pmatrix} \lambda_1 & 0 & 0 & \cdot & \cdot & 0 \\ 0 & \lambda_2 & 0 & \cdot & \cdot & 0 \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ 0 & 0 & 0 & \cdot & \cdot & \lambda_n \end{pmatrix}.$$

If we put

$$D = \begin{pmatrix} \lambda_1 & 0 & 0 & \cdot & \cdot & 0 \\ 0 & \lambda_2 & 0 & \cdot & \cdot & 0 \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ 0 & 0 & 0 & \cdot & \cdot & \lambda_n \end{pmatrix}$$

then we have

$$Q^T A Q = Q^{-1} A Q = D,$$

so that A is similar to D . Observe that

$$A = Q D Q^T.$$

We can use this observation to “uncouple” equations. Notice that for $x \in V^n$ we have

$$Ax = QDQ^T x,$$

$$Q^T(Ax) = D(Q^T x).$$

Suppose that $y = Ax$. If we make the change of variables

$$\hat{x} = Q^T x, \quad \hat{y} = Q^T y,$$

then we have

$$\hat{y} = D\hat{x}.$$

**Remark: FROM NOW ON ALL MATRICES, VECTORS,
AND SCALARS ARE ASSUMED TO BE REAL.**

Covariance Matrices

Suppose that

$$X = \begin{pmatrix} X_1 \\ X_2 \\ \cdot \\ \cdot \\ \cdot \\ X_n \end{pmatrix}$$

is a “random vector”. The *covariance matrix* $\mathbb{C}(X)$ of X is defined by

$$(\mathbb{C}(X))_{ij} = \text{Cov}(X_i, X_j) = \text{Cov}(X_j, X_i).$$

Observe that $\mathbb{C}(X)$ is symmetric. Recall that for $L \in M^{n \times n}$ we have

$$\mathbb{C}(LX) = L\mathbb{C}(X)L^T.$$

Let $\lambda_1, \lambda_2, \dots, \lambda_n$ be the eigenvalues of $\mathbb{C}(X)$ and $e^{(1)}, e^{(2)}, \dots, e^{(n)}$ be associated eigenvectors satisfying

$$(e^{(i)})^\top e^{(j)} = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases}.$$

Let Q be the $n \times n$ matrix whose i^{th} column is $e^{(i)}$, i.e.

$$Q_{ij} = e_i^{(j)}.$$

Let us put

$$D = Q^\top \mathbb{C}(X) Q,$$

and observe that D is diagonal. Now we have

$$\mathbb{C}(Q^\top X) = Q^\top \mathbb{C}(X) (Q^\top)^\top = Q^\top \mathbb{C}(X) Q = D.$$

Let $\hat{X} = Q^\top X$, and observe that $X = Q\hat{X}$. Then we have that $\hat{X}_1, \hat{X}_2, \dots, \hat{X}_n$ are uncorrelated.

This procedure is extremely useful in many situations in mathematical finance. Let's look at a simple example.

Example: Suppose that $X = \begin{pmatrix} X_1 \\ X_2 \end{pmatrix}$ and we are given that

$$\mathbb{C}(X) = A = \begin{pmatrix} 41 & -12 \\ -12 & 34 \end{pmatrix}.$$

We have

$$A - \lambda I = \begin{pmatrix} 41 - \lambda & -12 \\ -12 & 34 - \lambda \end{pmatrix}$$

$$\det(A - \lambda I) = (41 - \lambda)(34 - \lambda) - (-12)(-12)$$

$$= \lambda^2 - 75\lambda + 1,250$$

$$= (\lambda - 25)(\lambda - 50).$$

The eigenvalues of A are $\lambda_1 = 25$, $\lambda_2 = 50$. Let us find an eigenvector $e^{(1)}$ corresponding to $\lambda_1 = 25$. We put

$$e^{(1)} = \begin{pmatrix} \xi_1 \\ \xi_2 \end{pmatrix}.$$

This leads us to the system of equations

$$\begin{pmatrix} 41 - 25 & -12 \\ -12 & 34 - 25 \end{pmatrix} \begin{pmatrix} \xi_1 \\ \xi_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

Both equations say that $4\xi_1 - 3\xi_2 = 0$, i.e.

$$\xi_1 = \frac{3}{4}\xi_2.$$

In order to have $(e^{(1)})^\top e^{(1)} = 1$, we choose

$$\xi_1 = \frac{3}{5}, \quad \xi_2 = \frac{4}{5},$$

so we have

$$e^{(1)} = \frac{1}{5} \begin{pmatrix} 3 \\ 4 \end{pmatrix}.$$

We put $e^{(2)} = \begin{pmatrix} \eta_1 \\ \eta_2 \end{pmatrix}$. This leads us to the system of equations

$$\begin{pmatrix} 41 - 50 & -12 \\ -12 & 34 - 50 \end{pmatrix} \begin{pmatrix} \eta_1 \\ \eta_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

Both equations say that $3\eta_1 + 4\eta_2 = 0$, i.e.

$$\eta_1 = -\frac{4}{3}\eta_2.$$

In order to have $(e^{(2)})^\top e^{(2)} = 1$, we choose

$$\eta_1 = -\frac{4}{5}, \quad \eta_2 = \frac{3}{5},$$

so we have

$$e^{(2)} = \frac{1}{5} \begin{pmatrix} -4 \\ 3 \end{pmatrix}.$$

Notice that

$$(e^{(1)})^\top e^{(2)} = 0.$$

We have

$$Q = \frac{1}{5} \begin{pmatrix} 3 & -4 \\ 4 & 3 \end{pmatrix}, \quad Q^\top = \frac{1}{5} \begin{pmatrix} 3 & 4 \\ -4 & 3 \end{pmatrix}.$$

We see that

$$AQ = \frac{1}{5} \begin{pmatrix} 41 & -12 \\ -12 & 34 \end{pmatrix} \begin{pmatrix} 3 & -4 \\ 4 & 3 \end{pmatrix} = \begin{pmatrix} 15 & -40 \\ 20 & 30 \end{pmatrix}$$

$$Q^\top AQ = \frac{1}{5} \begin{pmatrix} 3 & 4 \\ -4 & 3 \end{pmatrix} \begin{pmatrix} 15 & -40 \\ 20 & 30 \end{pmatrix} = \begin{pmatrix} 25 & 0 \\ 0 & 50 \end{pmatrix}.$$

Let us put

$$\hat{X} = Q^T X = \frac{1}{5} \begin{pmatrix} 3 & 4 \\ -4 & 3 \end{pmatrix} \begin{pmatrix} X_1 \\ X_2 \end{pmatrix} = \frac{1}{5} \begin{pmatrix} 3X_1 + 4X_2 \\ -4X_1 + 3X_2 \end{pmatrix}.$$

We have

$$\hat{X}_1 = \frac{3}{5}X_1 + \frac{4}{5}X_2, \quad \hat{X}_2 = -\frac{4}{5}X_1 + \frac{3}{5}X_2.$$

Claim: $\text{Cov}(\hat{X}_1, \hat{X}_2) = 0$

Check: $\text{Cov}\left(\frac{3}{5}X_1 + \frac{4}{5}X_2, -\frac{4}{5}X_1 + \frac{3}{5}X_2\right)$

$$\begin{aligned} &= -\frac{12}{25}\text{Cov}(X_1, X_1) + \frac{9}{25}\text{Cov}(X_1, X_2) - \frac{16}{25}\text{Cov}(X_1, X_2) \\ &\quad + \frac{12}{25}\text{Cov}(X_2, X_2) = \frac{12}{25}(-41 - 9 + 16 + 34) = 0. \end{aligned}$$

Positive Definite and Positive Semidefinite Matrices

Let A be a real $n \times n$ matrix with $A^T = A$. We say that A is

(i) *positive definite* provided that

$$x^T A x > 0 \text{ for all } x \in V^n \text{ with } x \neq 0.$$

(ii) *positive semidefinite* provided that

$$x^T A x \geq 0 \text{ for all } x \in V^n.$$

Remark: (i) If A is positive definite then A is positive semidefinite.
(ii) Every covariance matrix is positive semidefinite.

Principal Minors

Let $A \in M^{n \times n}$ be given. For each $k = 1, 2, \dots, n$, we define $A_k \in M^{k \times k}$ by

$$(A_k)_{ij} = A_{ij}, \quad i, j = 1, 2, \dots, k.$$

Example: Let $A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}$. Then we have

$$A_1 = (1), \quad A_2 = \begin{pmatrix} 1 & 2 \\ 4 & 5 \end{pmatrix}, \quad A_3 = A.$$

Theorem: Assume that $A \in M^{n \times n}$ is real and $A^T = A$. The following 3 statements are equivalent:

- (i) A is positive definite.
- (ii) All eigenvalues of A are strictly positive.
- (iii) $\det(A_k) > 0$ for all $k = 1, 2, \dots, n$.

Theorem: Assume that $A \in M^{n \times n}$ is real and $A^T = A$. Then (i') and (ii') below are equivalent:

- (i') A is positive semidefinite.
- (ii') All eigenvalues of A are nonnegative.
- (iii') [There is no simple analog of (iii) that is equivalent to (i'), (ii').]

Example: Let $A = \begin{pmatrix} 5 & 1 & 2 \\ 1 & 2 & 3 \\ 2 & 3 & -8 \end{pmatrix}$. Observe that

$$\lambda_1 + \lambda_2 + \lambda_3 = \text{tr}(A) = 5 + 2 - 8 = -1 < 0.$$

At least one of the eigenvalues of A is strictly negative. We conclude that A is not positive semidefinite (and therefore is not positive definite). Observe that if we put $x = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$ then

$x^T A x = -8$. This argument can be used to show that the diagonal entries of a positive definite matrix must be strictly positive and the diagonal entries of a positive semidefinite matrix must be nonnegative.

Example: Let $A = \begin{pmatrix} 1 & 2 & 1 \\ 2 & 5 & 0 \\ 1 & 0 & 7 \end{pmatrix}$. You can easily check that

$$1 > 0, \quad \det \begin{pmatrix} 1 & 2 \\ 2 & 5 \end{pmatrix} = 1 > 0, \quad \det(A) = 2 > 0,$$

so A is positive definite.

Minimization and Convex Functions

Consider $f : V^n \rightarrow \mathbb{R}$, i.e.

$$f(x) = f \begin{pmatrix} x_1 \\ x_2 \\ \cdot \\ \cdot \\ x_n \end{pmatrix}.$$

We assume that f has continuous second order partial derivatives.
The gradient of f is given by

$$\nabla f(x) = \begin{pmatrix} \frac{\partial f}{\partial x_1}(x) \\ \cdot \\ \cdot \\ \frac{\partial f}{\partial x_n}(x) \end{pmatrix}$$

The Hessian matrix $H_f(x)$ is defined by

$$(H_f(x))_{ij} = \frac{\partial^2 f}{\partial x_i \partial x_j}(x) = \frac{\partial^2 f}{\partial x_j \partial x_i}(x).$$

Observe that $H_f(x)$ is symmetric. Taylor's Theorem for the first-order approximation with a second-order remainder says that for every $x, x^* \in V^n$ there exists z between x and x^* such that

$$\begin{aligned} f(x) = & f(x^*) + (\nabla f(x^*))^\top (x - x^*) \\ & + (x - x^*)^\top H_f(z)(x - x^*). \end{aligned}$$

f is said to be convex provided that

$$f(\alpha x + (1 - \alpha)y) \leq \alpha f(x) + (1 - \alpha)f(y)$$

for all $x, y \in V^n$ and all $\alpha \in [0, 1]$.

Theorem: f is convex if and only if $H_f(z)$ is positive semidefinite for every $z \in V^n$.

Theorem: Assume that f is convex and $x^* \in V^n$ satisfies $\nabla f(x^*) = 0$. Then

$$f(x) \geq f(x^*) \text{ for all } x \in V^n,$$

i.e. f attains a global minimum at x^* .

An Important Constrained Optimization Problem

Let $A \in M^{n \times n}$ be given and assume that A is real and that $A^T = A$. Put

$$S = \{x \in V^n : x^T x = 1\},$$

and define $f : V^n \rightarrow \mathbb{R}$ by

$$f(x) = x^T A x \text{ for all } x \in V^n.$$

Let $\lambda_1, \lambda_2, \dots, \lambda_n$ be the eigenvalues of A arranged so that

$$\lambda_1 \geq \lambda_2, \dots \geq \dots \geq \lambda_n.$$

We are interested in finding the maximum and minimum values of f on S .

Constrained Optimization Problem (Continued)

Theorem: Under the preceding assumptions, we have

- ▶ $\max\{f(x) : x \in S\} = \lambda_1$ and the maximum is attained at each $x \in S$ satisfying $Ax = \lambda_1 x$.
- ▶ $\min\{f(x) : x \in S\} = \lambda_n$ and the minimum is attained at each $x \in S$ satisfying $Ax = \lambda_n x$.

The proof is a straightforward application of the Lagrange Multiplier theory. Let's take a quick look at the main steps. Let

$$g(x) = x^T x = \|x\|^2 \quad \text{for all } x \in V^n.$$

Then $x \in S$ if and only if $g(x) = 1$. (Because we will need to differentiate g , it is convenient to use the square of the norm, rather than the norm, in the definition of g .)

Constrained Optimization Problem (Continued)

Since the set S is nonempty, closed, and bounded and the function f is continuous we know that f attains a maximum and a minimum on S . The Lagrange multiplier theorem tells us which vectors in S are possible candidates to be maximizers/minimizers. We simply need to check all of the candidates and see which ones provide the largest and smallest values for f .

If a maximum or minimum is attained at $x \in S$ then there is a scalar λ such that

$$\nabla f(x) = \lambda \nabla g(x).$$

You should convince yourself that

$$\nabla f(x) = 2Ax, \quad \nabla g(x) = 2x \quad \text{for all } x \in V^n.$$

If $x \in S$ is a maximizer or minimizer, then we must have

$$2Ax = 2\lambda x \quad \text{or} \quad Ax = \lambda x,$$

for some real number λ , i.e. x must be an eigenvector for A .

Orthogonality and Projections

$$\|x\| = \sqrt{x^\top x} \quad \text{for all } x \in V^n$$

$$\|x - y\| \quad \text{distance from } x \text{ to } y$$

$$x \cdot y = x^\top y = y^\top x = y \cdot x \quad \text{for all } x, y \in V^n$$

$\|x\|$ is called the *Euclidean norm* or *length* of x and $x \cdot y$ is called the *dot product* or *scalar product* of x with y . For these definitions, it is important that the vectors have real components.

Orthogonality and Projections (Continued)

Definition: Let $x, y \in V^n$ be given. We say that x and y are orthogonal if $x \cdot y = 0$.

Definition: Let $x^{(1)}, x^{(2)}, \dots, x^{(k)}$ be a list of vectors in V^n . We say that this list is

- (i) *orthogonal* provided that $x^{(i)} \cdot x^{(j)} = 0$ whenever $i \neq j$.
- (ii) *orthonormal* provided that the list is orthogonal and $\|x^{(i)}\| = 1$ for all $i = 1, 2, \dots, k$.

Remark: Observe that the list $u^{(1)}, u^{(2)}, \dots, u^{(k)} \in V^n$ is orthonormal if and only if

$$(u^{(i)})^\top u^{(j)} = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases}.$$

Orthogonality and Projections (Continued)

Proposition: Every orthonormal list of vectors is linearly independent.

Theorem: Let W be a subspace of V^n . Then there is an orthonormal list $u^{(1)}, u^{(2)}, \dots, u^{(k)}$ such that

$$\text{span}(u^{(1)}, u^{(2)}, \dots, u^{(n)}) = W.$$

Theorem: Let W be a subspace of V^n . There is exactly one matrix $P_W \in M^{n \times n}$ satisfying

- (i) $\mathcal{R}(P_W) = W$
- (ii) $\|P_W x - x\| \leq \|y - x\|$ for all $x \in V^n$ and $y \in W$. ($P_W x$ minimizes the distance between x and W .)

Facts about P_W :

1. $P_W x = x$ for all $x \in W$
2. $(P_W)^2 = P_W$
3. $(P_W)^\top = P_W$
4. $x - P_W x$ is orthogonal to y for every $y \in W$.

P_W is called the (orthogonal) projection matrix for W .

Gram-Schmidt Procedure

Given a linearly independent list of vectors

$x^{(1)}, x^{(2)}, \dots, x^{(k)} \in V^n$, it is possible to produce an orthonormal list $u^{(1)}, u^{(2)}, \dots, u^{(k)} \in V^n$ such that

$\text{span}(x^{(1)}, x^{(2)}, \dots, x^{(i)}) = \text{span}(u^{(1)}, u^{(2)}, \dots, u^{(i)})$ for all $i = 1, 2, \dots, k$.

Gram-Schmidt (Continued)

Here's how it works:

$$w^{(1)} = x^{(1)}, \quad u^{(1)} = \frac{w^{(1)}}{\|w^{(1)}\|},$$

$$w^{(2)} = x^{(2)} - (x^{(2)} \cdot u^{(1)})u^{(1)}, \quad u^{(2)} = \frac{w^{(2)}}{\|w^{(2)}\|}$$

$$w^{(3)} = x^{(3)} - (x^{(3)} \cdot u^{(1)})u^{(1)} - (x^{(3)} \cdot u^{(2)})u^{(2)}, \quad u^{(3)} = \frac{w^{(3)}}{\|w^{(3)}\|},$$

...

$$w^{(k)} = x^{(k)} - \sum_{i=1}^{k-1} (x^{(k)} \cdot u^{(i)})u^{(i)}, \quad u^{(k)} = \frac{w^{(k)}}{\|w^{(k)}\|}.$$

Constructing a Projection Matrix

Let W be a subspace of V^n and choose an orthonormal list of vectors $u^{(1)}, u^{(2)}, \dots, u^{(k)} \in V^n$ such that

$$\text{span}(u^{(1)}, u^{(2)}, \dots, u^{(k)}) = W.$$

Let A be the $n \times k$ matrix whose i^{th} column is $u^{(i)}$.

Claim: $P_W = AA^T$.

Example: $n = 3$, $k = 2$, $u^{(1)} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$, $u^{(2)} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$.

$$W = \{x \in V^3 : x_3 = 0\}$$

$$A = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix} \quad A^\top = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}$$

$$P_W = AA^\top = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

Example: $n = 3$, $k = 1$, $u^{(1)} = \frac{1}{\sqrt{3}} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$

$$W = \{x \in V^3 : x_1 = x_2 = x_3\}.$$

$$A = \frac{1}{\sqrt{3}} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \quad A^\top = \frac{1}{\sqrt{3}} \begin{pmatrix} 1 & 1 & 1 \end{pmatrix}$$

$$P_W = AA^\top = \frac{1}{3} \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}$$

Example (Continued)

Observe that

$$P_W x = \frac{1}{3} \begin{pmatrix} x_1 + x_2 + x_3 \\ x_1 + x_2 + x_3 \\ x_1 + x_2 + x_3 \end{pmatrix}.$$

Put $x = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$ and observe that

$$x - P_W x = \frac{1}{3} \begin{pmatrix} 2 \\ -1 \\ -1 \end{pmatrix}$$

is orthogonal to every vector in W .

Remark: For every subspace W of V^n and all vectors $x \in V^n$, $y \in W$ we have

$$(x - P_W x) \cdot y = 0.$$

Triangular Matrices and Decompositions

The system of equations $Tx = b$ is very easy to solve if $T \in M^{n \times n}$ is triangular and all diagonal entries are different from zero. One can use a simple step-by-step procedure to solve for the components of x one at a time. For example, suppose that $L \in M^{n \times n}$ is lower triangular and all diagonal entries are nonzero. The solution to $Lx = b$ can be found easily by observing that the first component equation is

$$L_{11}x_1 = b_1, \text{ so that } x_1 = \frac{b_1}{L_{11}}$$

and then the second equation becomes

$$L_{21}x_1 + L_{22}x_2 = b_2 \text{ so that } x_2 = \frac{b_2 - L_{21}x_1}{L_{22}}.$$

As we proceed one row at a time from top to bottom, there is only one unknown at each step. The situation is similar for upper triangular matrices.

L-U Decompositions

Let $A \in M^{n \times n}$ be given. A decomposition of A into a product of the form

$$(*) \quad A = LU,$$

where L is lower triangular and U is upper triangular is called an $L - U$ decomposition of A . Such a decomposition can be extremely useful. Suppose that $(*)$ holds and $b \in V^n$. Then we can solve $Ax = b$ by solving the pair of triangular systems

$$Ly = b, \quad Ux = y.$$

Cholesky Decomposition

A special type of $L - U$ decomposition, called a *Cholesky decomposition* will be important in several places MSCF.

Theorem: (Cholesky Decomposition) Let $A \in M^{n \times n}$ be given. Assume that A has real entries, $A^T = A$, and that A is positive definite. Then there exists exactly one lower triangular $L \in M^{n \times n}$ with positive entries on the diagonal such that

$$A = LL^T.$$

If A is symmetric and positive semidefinite, then it still has a Cholesky decomposition provided we allow some diagonal elements to be 0. In this case, the matrix L is not necessarily unique.

Example: $A = \begin{pmatrix} 1 & 2 \\ 2 & 5 \end{pmatrix}$

Since $1 > 0$ and $1 \cdot 5 - 2 \cdot 2 > 0$, we see that A is positive definite. We want to find a, b, c with $a, c > 0$ such that

$$A = \begin{pmatrix} a & 0 \\ b & c \end{pmatrix} \begin{pmatrix} a & b \\ 0 & c \end{pmatrix} = \begin{pmatrix} a^2 & ab \\ ab & b^2 + c^2 \end{pmatrix}$$

We find that $a = 1$, $b = 2$, $4 + c^2 = 5$, so $c = 1$ and

$$L = \begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix}.$$

Observe that

$$\begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 2 \\ 2 & 5 \end{pmatrix}.$$

General Cholesky Algorithm

- ▶ L_{k-1} denotes the upper left $(k-1) \times (k-1)$ block of L
- ▶ a_k denotes the first $k-1$ entries in column k of A
- ▶ l_k denotes the first $k-1$ entries in column k of $L^\top$

Then we have

$$L_1 = \sqrt{A_{11}}, \quad \text{and}$$

for $k = 2, 3, \dots, n$ we solve

$$L_{k-1} l_k = a_k \quad \text{for } l_k$$

and we put

$$L_{kk} = \sqrt{A_{kk} - (l_k)^\top l_k}, \quad L_k = \begin{pmatrix} L_{k-1} & 0 \\ l_k^\top & L_{kk} \end{pmatrix}.$$

Example

Let

$$A = \begin{pmatrix} 25 & 15 & -5 \\ 15 & 18 & 0 \\ -5 & 0 & 11 \end{pmatrix}.$$

You should check as an exercise that A is positive definite. Let's find the Cholesky decomposition using the algorithm.

$$L_1 = \sqrt{25} = 5.$$

$$L_1 l_2 = a_2, \quad 5l_2 = 15, \quad \text{so } l_2 = 3.$$

$$L_{22} = \sqrt{18 - (3)3} = 3, \quad \text{so}$$

$$L_2 = \begin{pmatrix} 5 & 0 \\ 3 & 3 \end{pmatrix}.$$

Example Continued

$$L_2 l_3 = a_3, \quad \begin{pmatrix} 5 & 0 \\ 3 & 3 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -5 \\ 0 \end{pmatrix}$$

$$x = -1, \quad y = 1, \quad \text{so } l_3 = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$$

$$L_{33} = \sqrt{A_{33} - l_3^\top l_3} = \sqrt{11 - 2} = 3.$$

$$L_3 = \begin{pmatrix} 5 & 0 & 0 \\ 3 & 3 & 0 \\ -1 & 1 & 3 \end{pmatrix}$$

Cholesky Decomposition Exercise

Let

$$A = \begin{pmatrix} 16 & 4 & 4 & -4 \\ 4 & 10 & 4 & 2 \\ 4 & 4 & 6 & -2 \\ -4 & 2 & -2 & 4 \end{pmatrix}.$$

Verify that

$$L = \begin{pmatrix} 4 & 0 & 0 & 0 \\ 1 & 3 & 0 & 0 \\ 1 & 1 & 2 & 0 \\ -1 & 1 & -1 & 1 \end{pmatrix}.$$